

# Exponential Expressions

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## MA.7.NSO.1.1

I can apply more than one law of exponents to evaluate an expression and to write an equivalent one.

# Warm-Up

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1. Find the value:  $\frac{10^5}{10^3}$
2. Is  $\sqrt{50}$  closer to 7 or to 8?
3. Solve  $4(x - 2) = 12$
4. Evaluate  $2 + 3 \cdot 4^2$

# Warm-Up

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1. Find the value:  $\frac{10^5}{10^3}$  **100**
2. Is  $\sqrt{50}$  closer to 7 or to 8? **7**
3. Solve  $4(x - 2) = 12$   **$x = 5$**
4. Evaluate  $2 + 3 \cdot 4^2$  **50**

# Notes

## I. The seven laws, in one place

| Law                 | Rule                          | Example                             |
|---------------------|-------------------------------|-------------------------------------|
| Product of powers   | $a^m \cdot a^n = a^{m+n}$     | $2^3 \cdot 2^1 = 2^4$               |
| Quotient of powers  | $a^m \div a^n = a^{m-n}$      | $(-3)^7 \div (-3)^2 = (-3)^5$       |
| Power of a product  | $(ab)^m = a^m \cdot b^m$      | $(-7 \cdot 3)^2 = (-7)^2 \cdot 3^2$ |
| Power of a quotient | $(a \div b)^m = a^m \div b^m$ | $(13 \div 5)^3 = 13^3 \div 5^3$     |
| Power of a power    | $(a^m)^n = a^{mn}$            | $(6^3)^4 = 6^{12}$                  |
| Identity exponent   | $a^1 = a$                     | $15^1 = 15$                         |
| Zero exponent       | $a^0 = 1$                     | $127^0 = 1$                         |

# Notes

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## II. More than one law — make a plan, then follow it

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$$\text{Winston: } \left( \frac{2^6}{2^4} \right)^3 = (2^2)^3 = 2^6 = 64$$

$$\text{Yvette: } \left( \frac{2^6}{2^4} \right)^3 = \frac{(2^6)^3}{(2^4)^3} = \frac{2^{18}}{2^{12}} = 2^6 = 64$$

Different plans, same value. Any correct sequence of laws gives the same answer.

# Notes

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## III. Order of operations still applies

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- A. Parentheses first — including a fraction bar, which is a pair of parentheses in disguise.
- B. Exponents next — the laws of exponents are applied here.
- C. Then multiply and divide, left to right; then add and subtract.
- D. **Never across a + or a -:**  $2^3 + 2^2 = 8 + 4 = 12$ , not  $2^5$ .

# Example 1

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$$\frac{4^5 \cdot 4^0}{4^2}$$

**Make a plan. Then find the value.**

# Example 1

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$$\frac{4^5 \cdot 4^0}{4^2} = \frac{4^{5+0}}{4^2} = \frac{4^5}{4^2}$$

$$\frac{4^5 \cdot 4^0}{4^2} = \frac{4^5 \cdot 1}{4^2} = \frac{4^5}{4^2}$$

$$\frac{4^5}{4^2} = 4^3 = 64$$

**Answer: 64**

# Your Turn

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$$\frac{5^4 \cdot 5^0}{5^2}$$

**Make a plan. Then find the value.**

# Your Turn

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$$\frac{5^4 \cdot 5^0}{5^2}$$

**Make a plan. Then find the value.**

$$5^0 = 1, \text{ so the top is } 5^4 \cdot 1 = 5^4$$

$$\frac{5^4}{5^2} = 5^{4-2} = 5^2 = 25$$

**Answer:  $5^2 = 25$**

## Example 2

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$$11.6^0 \cdot \frac{(-15)^4}{1.5^4}$$

**Make a plan. Then find the value.**

## Example 2

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$$11.6^0 \cdot \frac{(-15)^4}{1.5^4} = 1 \cdot \left(\frac{-15}{1.5}\right)^4$$

$$\left(\frac{-15}{1.5}\right)^4 = (-10)^4$$

$$(-10)^4 = 10000$$

**Answer: 10,000**

# Your Turn

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$$7^0 \cdot \frac{(-12)^3}{4^3}$$

**Make a plan. Then find the value.**

# Your Turn

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$$7^0 \cdot \frac{(-12)^3}{4^3}$$

**Make a plan. Then find the value.**

$$7^0 = 1$$

$$\frac{(-12)^3}{4^3} = \left(\frac{-12}{4}\right)^3 = (-3)^3 = -27$$

**Answer: -27**

# Whiteboards • Question 1 of 9

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A warehouse holds  $6^4$  boxes in  $6^2$  equal rows.

Each box weighs 3 kilograms.

**How many boxes are in each row?**

**Answer it.**

# Whiteboards • Question 1 of 9

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A warehouse holds  $6^4$  boxes in  $6^2$  equal rows.

Each box weighs 3 kilograms.

**How many boxes are in each row?**

$$6^4 \div 6^2 = 6^{4-2} = 6^2$$

$$6^2 = 36$$

**Answer: 36**

# Whiteboards · Question 2 of 9

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$$(3^0 \cdot 3^2)^3$$

**Answer it.**

# Whiteboards • Question 2 of 9

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$$(3^0 \cdot 3^2)^3$$

Inside first:  $3^0 \cdot 3^2 = 3^{0+2} = 3^2$

$$(3^2)^3 = 3^{2 \cdot 3} = 3^6 = 729$$

**Answer: 729**

# Whiteboards • Question 3 of 9

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$$\frac{(0.5^3)^2}{(0.5)^2}$$

**Answer it.**

# Whiteboards • Question 3 of 9

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$$\frac{(0.5^3)^2}{(0.5)^2}$$

$$\text{Top: } (0.5^3)^2 = 0.5^{3 \cdot 2} = 0.5^6$$

$$\frac{0.5^6}{0.5^2} = 0.5^{6-2} = 0.5^4 = 0.0625$$

**Answer: 0.0625**

# Whiteboards · Question 4 of 9

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$$\left(\frac{2.5^5}{2.5^3}\right)^1$$

**Answer it.**

# Whiteboards • Question 4 of 9

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$$\left( \frac{2.5^5}{2.5^3} \right)^1$$

$$\frac{2.5^5}{2.5^3} = 2.5^{5-3} = 2.5^2$$

$$(2.5^2)^1 = 2.5^2 = 6.25$$

**Answer: 6.25**

# Whiteboards • Question 5 of 9

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$$\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$$

Which is the value?

**A.** 1/4

**B.**  $2^{22}$

**C.** 0

**D.** 4

**Answer it.**

# Whiteboards • Question 5 of 9

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$$\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$$

Which is the value?

$$3 \cdot 2^0 = 1, \text{ so the top is } (2^2)^6 = 2^{2 \cdot 6} = 2^{12}$$

$$\frac{2^{12}}{2^{10}} = 2^{12-10} = 2^2 = 4$$

**Answer: D — 4**

# Whiteboards • Question 6 of 9

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$$\frac{0.2^3}{2^3} \cdot 0.1^2$$

Which expression is equivalent?

**A.**  $0.2^0 \cdot 0.1^2$

**B.**  $0.2^3 \cdot 0.1^2$

**C.**  $0.1^5$

**D.**  $0.1^6$

**Answer it.**

# Whiteboards • Question 6 of 9

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$$\frac{0.2^3}{2^3} \cdot 0.1^2$$

Which expression is equivalent?

$$\frac{0.2^3}{2^3} = \left(\frac{0.2}{2}\right)^3 = 0.1^3$$

$$0.1^3 \cdot 0.1^2 = 0.1^{3+2} = 0.1^5$$

**Answer: C —  $0.1^5$**

# Whiteboards • Question 7 of 9

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$$0.1^2 \cdot (0.1^1)^2$$

**Answer it.**

# Whiteboards · Question 7 of 9

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$$0.1^2 \cdot (0.1^1)^2$$

$$(0.1^1)^2 = 0.1^{1 \cdot 2} = 0.1^2$$

$$0.1^2 \cdot 0.1^2 = 0.1^{2+2} = 0.1^4 = 0.0001$$

**Answer: 0.0001**

# Whiteboards • Question 8 of 9

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$$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

**Answer it.**

# Whiteboards • Question 8 of 9

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$$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

Top:  $(-4)^2 \cdot (-4)^1 = (-4)^{2+1} = (-4)^3$

Bottom:  $(-4)^0 = 1$

$$(-4)^3 = -64$$

**Answer: -64**

## Whiteboards • Question 9 of 9

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Winston writes  $\left(\frac{4^7}{4^4}\right)^2 = (4^3)^2$ . Yvette writes  $\left(\frac{4^7}{4^4}\right)^2 = \frac{4^{14}}{4^8}$ .

Do they get the same value? Explain.

**Write your answer in sentences.**

# Whiteboards • Question 9 of 9

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Winston writes  $\left(\frac{4^7}{4^4}\right)^2 = (4^3)^2$ . Yvette writes  $\left(\frac{4^7}{4^4}\right)^2 = \frac{4^{14}}{4^8}$ .

Do they get the same value? Explain.

Winston:  $(4^3)^2 = 4^{3 \cdot 2} = 4^6$

Yvette:  $\frac{4^{14}}{4^8} = 4^{14-8} = 4^6$

$$4^6 = 4,096$$

**Answer: Yes — both give  $4^6 = 4,096$**

# Independent Set

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1. Find the value.  $\frac{4^2 \cdot 4^0}{4^1}$
2. Find the value.  $(2^0 \cdot 2^3)^2$
3. Write as one power.  $\left(\frac{5^8}{5^5}\right)^3$
4. Find the value.  $0.2^2 \cdot (0.2^1)^2$
5. Find the value.  $\frac{(-3)^3 \cdot (-3)^0}{(-3)^2}$
6. Show that  $\left(\frac{10^4}{10^2}\right)^2$  and  $\frac{10^8}{10^4}$  have the same value.

# IXL

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1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Evaluate expressions using properties of exponents — ZE9